

Privacy-Preserving Integrity Proofs for Student-Society Voting-Adjacent Workflows: A Phase C Pilot of Project Simurgh at Macquarie University

Mohammad Raouf Abedini

Department of Computing

Macquarie University

Sydney, Australia

mohammadraouf.abedini@students.mq.edu.au

Abstract—We show that a consent-based shadow-mode system can collect session-integrity metadata alongside a real voting event without recording ballot choices, enforced structurally rather than by policy alone. Digital voting tools raise an integrity question that existing cryptographic protocols (end-to-end verifiable voting, mix-nets) do not address: can a participant verify their session was conducted fairly without revealing their vote? Project Simurgh answers this with an HMAC audit chain, client-side ballot-field discard before each network call, and a server-side forbidden-field guard that rejects any ballot-choice key. A 30-session Phase C pilot (31 consented, 1 withdrawn) confirmed `ballot_choice_recorded_by_simurgh: false` in every submitted session; 359/359 automated tests, 8/8 smoke gates, and 10/10 security-audit gates passed at closeout. After data collection ended, all write endpoints returned HTTP 410 Gone under a server-side collection-closed flag, verifiable by any caller able to reach the pilot API. This study is voting-adjacent: it does not implement ballot cryptography, voter eligibility verification, coercion resistance, tally protection, or public-election certification.

Index Terms—privacy-preserving integrity, shadow-mode pilot, student-society voting, HMAC audit chain, consent-based telemetry, collection closure

I. INTRODUCTION

Student-society governance increasingly relies on digital voting tools for event preference polls, committee elections, and budget decisions. These tools leave a question open that existing cryptographic solutions do not answer: can a participant verify their session was handled honestly (no screen capture, no ballot recording, no content-level surveillance) without revealing their choice?

End-to-end verifiable voting schemes [1], [2] solve the *outcome* integrity problem (did the tally reflect all ballots?) but do not address *session* integrity (was the participant’s interaction free of covert collection?). Project Simurgh is a research prototype that targets session integrity through

privacy-preserving metadata collection and tamper-evident audit chains [3]. It was developed in response to the Invisible Window attack [4], which demonstrated that OS-level display affinity APIs allow application windows to evade browser-based screen capture entirely, undermining the display-fidelity assumption on which WebRTC proctoring relies.

This paper reports a Phase C pilot in which Project Simurgh ran in shadow mode alongside a real MQ Persian Society event preference poll, collecting session-level integrity metadata in parallel with no effect on the official result. The pilot was not a study of election security, voting-system integrity, or electoral fraud prevention.

Contributions:

- A consent-to-submit pilot with 30 submitted sessions confirming `ballot_choice_recorded_by_simurgh: false` in every session, backed by smoke, security-audit, and privacy-audit gate suites (Sec. V).
- A server-side collection-closure mechanism returning HTTP 410 Gone on all write routes, enforceable and verifiable independently of the client UI (Sec. III).
- A reproducible evidence baseline at tag `v0.5.0-voting-pilot-phase-c-closeout` for future work on privacy-preserving integrity proofs in voting-adjacent contexts.

II. RELATED WORK

A. End-to-End Verifiable Voting

Systems such as Helios [1] use homomorphic encryption or mix-nets so voters can verify their encrypted ballot was counted without revealing the plaintext choice. Civitas [2] adds coercion resistance; STAR-Vote [5] adds risk-limiting-audit integration.

Project Simurgh differs in scope. It does not implement ballot encryption, voter-verifiable tallying, or public election certification. Where end-to-end verifiable voting proves the integrity of the *outcome*, Project Simurgh studies the integrity

Voting-adjacent case study, not a production election-security claim. Research prototype only. Source: <https://github.com/Raouf128/Project-Simurgh>. Tag: `v0.5.0-voting-pilot-phase-c-closeout`.

of the *participation session*: whether the system collected only what it disclosed and nothing more.

B. Remote and Internet Voting Security

Remote electronic voting introduces risks beyond ordinary web application security: endpoint compromise, coercion, eligibility verification, ballot secrecy, and tally integrity. NIST guidance [6] identifies threat categories for remote electronic voting and recommends caution around internet-connected election infrastructure. The National Academies report [7] emphasizes that vote collection, alteration, destruction, counting, and reporting each require separate mitigations.

This pilot addresses none of those concerns. It is a student-society research case study, not a production online-voting system. It does not address eligibility, coercion resistance, malware-compromised endpoints, official tally integrity, or public-election certification.

C. Voting Standards, Privacy, and Auditability

The U.S. Election Assistance Commission adopted VVSG 2.0 in 2021 [8], establishing updated requirements for privacy, accessibility, auditability, and software independence in voting systems. Modern voting-system guidance consistently requires audit records that do not reveal individual ballot choices.

Project Simurgh aligns with this direction: the HMAC audit chain records session lifecycle events without ballot content, but remains outside public-election certification scope.

D. Australian Technology-Assisted Voting

Several Australian jurisdictions have piloted technology-assisted voting, including iVote in New South Wales. The NSW Electoral Commission’s technology-assisted voting review [9] examined privacy, accessibility, and security trade-offs, recommending careful scoping and ongoing evaluation before broader deployment.

Australia’s cautious posture toward technology-assisted voting makes small, bounded pilots valuable for studying consent, privacy boundaries, and session-integrity evidence without affecting official outcomes.

E. Privacy-Preserving Telemetry and Data Minimization

Project Simurgh’s ballot-field exclusion architecture follows data-minimization principles [10], [11]: the system collects the minimum metadata required for a session-integrity signal and structurally prevents ballot-choice data from reaching the server. Data that is never collected cannot be leaked.

F. Position of This Work

This paper contributes a bounded empirical evaluation of privacy-preserving integrity evidence around a voting-adjacent workflow. The system remains blind to ballot choice and does not discipline or flag participants. The contribution is not a new voting protocol; it is evidence that shadow-mode session-integrity collection is achievable at student-governance scale.

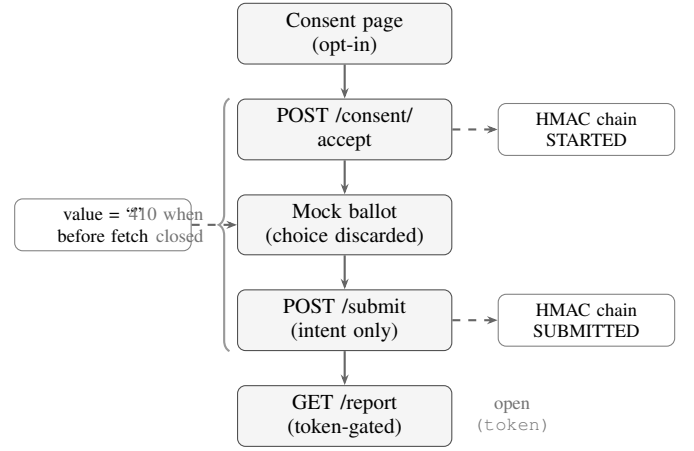


Fig. 1. Phase C session lifecycle. Ballot choice is discarded client-side before the submit call (left). HMAC chain events append on consent and submit (right). Under the collection-closed flag, write routes return HTTP 410; the report endpoint remains token-protected.

III. SYSTEM DESIGN

Fig. 1 shows the Phase C session lifecycle.

A. Consent Flow

The consent page disclosed two categories: data Project Simurgh may collect across its deployments (session timestamps, focus-loss and paste counts, audit-chain validity signal, and an anonymous HMAC-hashed session code) and data it never collects (ballot choices, screen recordings, webcam/audio, typed or pasted content, raw process names, window titles, or device serial identifiers). In Phase C, implemented collection comprised session timestamps, audit-chain lifecycle events, and an anonymous participant code; focus-loss and paste counts are exam-integrity features of the broader Project Simurgh system and were not active in the voting pilot. Participation was voluntary; declining produced no consequence.

On consent, the server issued an anonymous participant code and a signed session token. No personally identifying information was collected at any step.

B. Ballot-Field Exclusion

Before the submit `fetch` call, the submit page set all radio button `value` attributes to an empty string, so the ballot field was structurally absent from the request body. (The consent `fetch` occurs before any ballot interaction and carries no ballot field.) The server provided a second layer: the `FORBIDDEN_BALLOT_FIELDS` set rejected any submission body containing a known ballot-choice key with HTTP 400, with no echo of the supplied value.

C. HMAC Audit Chain

Each session maintained an append-only HMAC audit chain recording lifecycle events: `STARTED`, `SUBMITTED`, and optionally `WITHDRAWN` or `REPORT_EXPORTED`. The chain initialized on consent; the server verified its integrity on report export.

D. Collection-Closure Posture

After reaching the 30-session target, we set the environment variable `SIMURGH_VOTING_PILOT_COLLECTION_CLOSED=true`. With this flag active:

- `POST /api/voting-pilot/consent/accept` returns 410 Gone.
- `POST /api/voting-pilot/submit` returns 410 Gone.
- `POST /api/voting-pilot/withdraw` returns 410 Gone.
- `GET /api/voting-pilot/:id/report` remains available, token-protected.

The closure middleware fires before authentication on write routes; a valid session token cannot bypass it.

IV. PHASE C PILOT

A. Study Design

The pilot ran in shadow mode alongside the MQ Persian Society 2026 event preference poll, asking members to select one of four proposed social events (Nowruz cultural night, Persian movie night, career/networking night, food and music night). Project Simurgh ran in parallel without access to or influence on the official ballot-counting system. Each participant who joined the pilot completed two steps: a consent page explaining data collection and a mock ballot page mirroring the official event poll. The mock ballot choice was discarded in the browser before any network call; Project Simurgh received only a submit-intent signal.

Participants were members of the society. Participation was voluntary and governed by the Phase C participant notice and data management addendum approved by the society executive on 2026-06-04. No incentive was offered.

B. Session Counts

Of 31 sessions that accepted the consent flow, 30 completed the submit step and formed the primary analysis dataset. The participant in one session (`vp_4fcc741a`) withdrew before submitting; that session is excluded from all analysis and its report endpoint returns HTTP 403.

C. Governance and Ethics

The pilot proceeded after society executive approval, a locked participant notice, and a data management addendum. Because the pilot collected no ballot choices, no names, no student identifiers, no email addresses, and no content-level data, the study reports only aggregate technical outcomes. No individual-level analysis, disciplinary action, or re-identification was performed. Participation was fully voluntary with no consequence for declining or withdrawing. This paper does not claim formal institutional human-research ethics approval; it reports a society-approved internal pilot with aggregate technical outcomes only.

TABLE I
SESSION COUNTS

Metric	Count
Consented pilot sessions	31
Submitted sessions (primary analysis)	30
Withdrawn sessions	1
Official vote impact	0
Ballot choices collected by Project Simurgh	0

TABLE II
PRIVACY ASSERTIONS — ALL 30 SUBMITTED SESSIONS

Assertion	Result
<code>ballot_choice_recorded</code>	false
Ballot choice transmitted to server	No
Screen capture	No
Webcam / audio	No
Typed content	No
Pasted content	No
Personal device identifiers	No
Raw process names	No
Raw window titles	No
<code>official_vote_impact</code>	false
Privacy audit (<code>privacy-audit.mjs</code>)	PASS

D. Data Management

The server held live session data in memory only. The application persisted no live pilot session records; aggregate gate evidence and closeout artifacts were separately archived in the repository. The store reset on server restart, consistent with the data management addendum, which specified no persistent storage of individual pilot session records.

V. RESULTS

A. Dataset

Phase C recorded 31 consented pilot sessions. Thirty sessions completed the submit step and formed the primary analysis dataset; one session was withdrawn before submission and excluded from report export and analysis. Table I summarizes the retained dataset.

B. Privacy Assertions

Row 1 abbreviates the full field name `ballot_choice_recorded_by_simurgh` for column width.

The `ballot_choice_recorded_by_simurgh` field is a behavioral guarantee enforced by client-side discard and server-side forbidden-field rejection. Because the session store was in-memory and cleared after collection, individual session reports are not available post-hoc; the automated smoke suite verifies the guarantee at every gate run.

C. Integrity Flow

All 30 submitted sessions completed the consent-to-submit flow. The HMAC audit chain recorded `STARTED` on consent and `SUBMITTED` on submit for each. The withdrawn session's chain recorded `WITHDRAWN`; its report endpoint returned HTTP 403, confirming exclusion from analysis.

TABLE III
SAFETY GATES — CLOSEOUT BASELINE

Gate	Result
Node tests (npm test)	359/359 pass
npm audit (--audit-level=high)	0 high/critical
Privacy audit	PASS (52 evidence files)
smoke-voting-pilot.sh	8/8 pass
security-audit-voting-pilot.sh	10/10 pass
smoke-voting-pilot-closed.sh	5/5 pass

D. Safety Gates

Gate evidence files are archived in the repository under the Phase C closeout evidence directory at tag `v0.5.0-voting-pilot-phase-c-closeout`.

E. Collection Closure

After collection reached 30 sessions, all write endpoints returned HTTP 410 Gone. The closure smoke suite (5/5 gates) confirmed that consent/accept, submit, and withdraw are unreachable under the closed configuration, while report export remains available with a valid token.

VI. DISCUSSION

A. Interpretation

The pilot shows that a consent-based shadow-mode prototype can operate alongside a real student-society voting event without collecting ballot choices, content-level data, screen or audio/video recordings, raw local identifiers, or personal device identifiers. The privacy assertions in Table II hold by system design: the ballot field is structurally absent from the server request body in every session, not because participants complied with a policy.

The server-side collection-closure mechanism produces a reproducible, auditable end-of-collection posture. Any caller able to reach the pilot API can verify that write routes return HTTP 410 under the closed configuration; the mechanism does not depend on the client UI.

B. Limitations

- The 30-session dataset is a proof-of-concept pilot. No statistical power calculation was performed; results should not be generalized beyond this scope.
- Sessions were anonymous; no demographic or behavioral analysis of participants is possible or intended.
- The pilot ran in shadow mode only. No claim extends to a non-shadow deployment.
- The in-memory store means session data did not survive server restarts. This suits a research prototype but is not a production data-management posture.
- `ballot_choice_recorded_by_simurgh: false` is a behavioral guarantee verified by the automated gate suite; per-session audit reports are not available post-hoc due to in-memory storage.

C. Non-Claims

This pilot does not claim to:

- Secure or audit the official MQ Persian Society election result.
- Detect or prevent election fraud.
- Provide hardware-rooted attestation.
- Replace or improve the society’s own voting system.
- Generalize to national, governmental, or binding elections.

VII. CONCLUSION

Thirty sessions completed the full consent-to-submit flow in Phase C of Project Simurgh. In every session, the system collected no ballot choice, screen recording, webcam or audio data, typed content, pasted content, or personal device identifier. After collection closed, all write endpoints returned HTTP 410 Gone under a server-side flag, verifiable by any caller able to reach the pilot API, confirming that collection closure is enforced at the API layer, not only at the UI.

This pilot demonstrates that privacy-preserving session-integrity evidence is structurally achievable alongside a real student-society voting event, without access to or influence on the official vote.

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